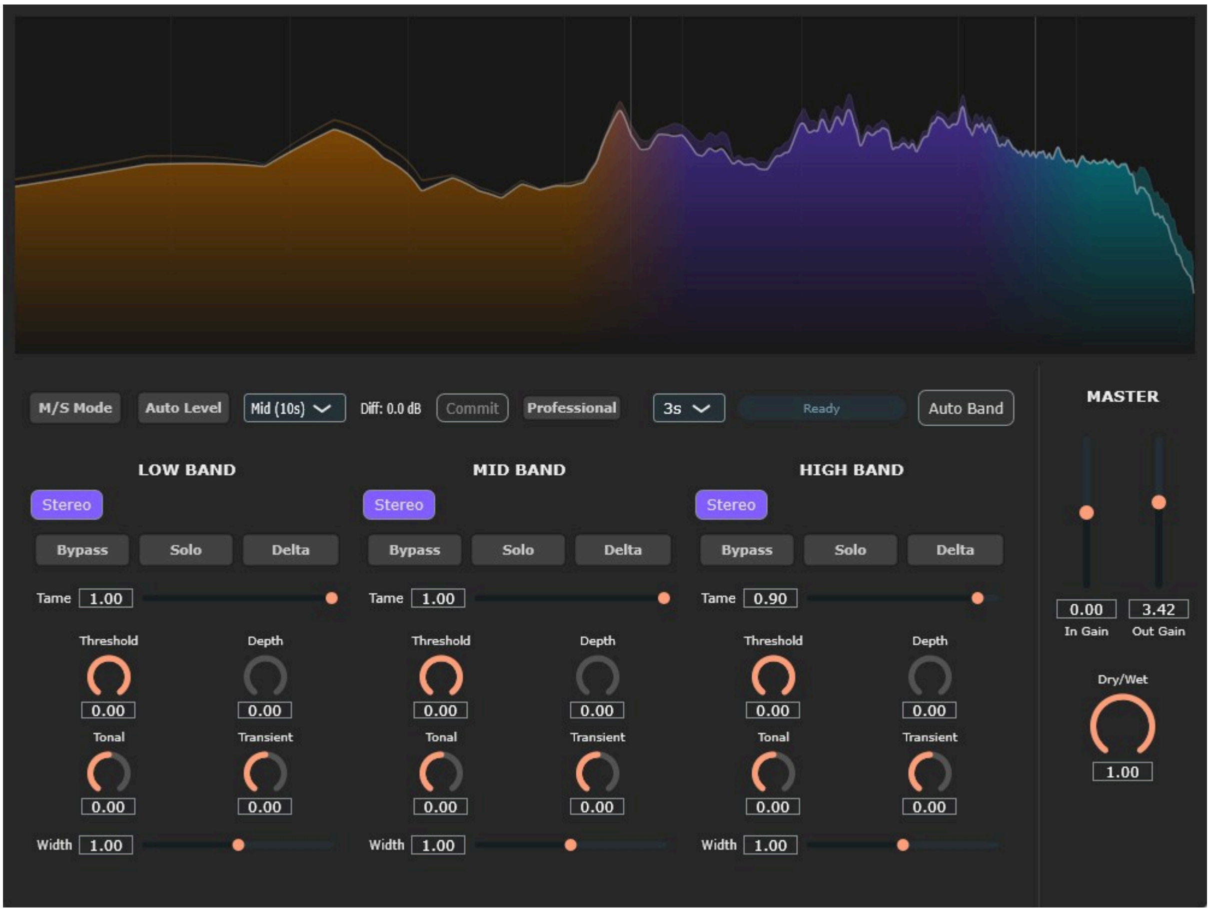
LUMINA User Manual



1. Plugin Overview and Signal Flow

LUMINA is a next-generation spectral dynamics processor that, instead of suppressing the "overall volume" like a conventional compressor, uses high-resolution STFT (Kaiser-Bessel WOLA engine) to decompose sound into 512 frequency "bins" and pinpointedly resolves "jarring frequency clashes" based on a psychoacoustic masking model.🔄 **Strict Signal Flow (Internal Processing Order)**

To prevent phase degradation and latency increase, LUMINA clearly separates the processing into "**simultaneous application in the frequency domain (single pass)**" and "**safe spatial control in the time domain**."**【Phase 1】Pre-processing in the Time Domain**

1. **Input Gain:** Applies the master Input Gain to the input signal (zipper noise is eliminated by juce::SmoothedValue).
2. **Oversampling (Up):** Upsamples up to 4x to prevent aliasing noise.
3. **M/S Encode:** If M/S Mode is ON, the signal is separated into Mid and Side components.

【Phase 2】Analysis and Simultaneous Application in the Frequency Domain (Single Pass)

STFT decomposes the waveform into a spectrum (amplitude). The following processes A–C are **calculated in parallel based on the "unprocessed original sound**."

- **Process A: Tame (Resonance Suppression)**
- Detects peaks protruding from the envelope (a contour smoothed by TPT) and calculates the suppression gain (tGain).
- **Process B: Threshold / Depth (Masking Suppression)**
- Uses the Bark scale model to detect "unnecessary muddiness that masks other sounds" and calculates the suppression gain (pureRed).
- **Process C: Tonal / Transient (Sustained/Percussive Sound Modulation)**
- Calculates the ratio of sustained/percussive sounds using HPSS technology and uses this as a modulator to dynamically increase or decrease the intensity of Process A and B, as well as the basic volume of the band.

🔥 **Simultaneous Application (Total Gain Multiplication):**

The calculated tGain and dGain (Threshold portion) are multiplied together to form a single "Final Gain." This is **multiplied only once and simultaneously across all 512 frequency bins**. Since this is not a serial process, sound quality degradation does not occur.**【Phase 3】Post-processing and Spatial Control in the Time Domain**

Inverse FFT (IFFT) is used to convert the processed spectrum back into an audio waveform.

1. **M/S Decode:** Converts back to L/R waveforms.
2. 🌌 **Width Processing (TPT Crossover & Decorrelation):**
3. *Because manipulating phase within the STFT causes unpleasant artifacts, stereo expansion is performed using safe filters in the time domain.*
    - Audio is split into Low / Mid / High using ZDF (Zero-Delay Feedback) structured TPT Linkwitz-Riley filters.
    - Different Width algorithms are applied to each band (details below), and the signals are then recombined.
4. **Oversampling (Down):** Returns to the original sample rate.
5. **Auto Level & Output:** Automatically corrects the RMS level difference before and after processing, and applies the Output Gain for output.

-----2. Master & Global Controls

- **Input / Output Gain (-24dB 〜 +24dB):**
- Adjusts the gain for the input and final output.
- **Dry / Wet (0.0 〜 1.0):**
- The mix balance between the original and processed sound. Internal latency (including lookahead) is completely automatically compensated.
- **M/S Mode (Toggle):**
- Switches whether stereo signal dynamics processing is performed independently on Mid/Side components or on L/R.

- **Auto Band (Trigger):**
- Analyzes the "spectral valleys (bands with low energy)" of the input signal for a specified time and automatically sets the optimal crossover frequencies.
- **Crossover 1 / 2 (20Hz 〜 20kHz):**
- The boundary frequencies for Low/Mid and Mid/High. Thanks to the TPT filter, they do not diverge (create noise) even when moved drastically during playback.
- **Auto Level (Toggle / Commit):**
- When ON, it automatically compensates for volume changes before and after processing. Pressing "Commit" finalizes this correction amount by embedding it into the Output Gain.

-----3. Band Section (Low / Mid / High)🎛️ Differentiating Tame and Threshold / Depth

These two are not "serial" but are independent logics that act simultaneously at the same point.

- **Tame (0.0 〜 1.0):**
  - **Role:** Smoothly shaves off "sudden peaks and resonances" occurring in each FFT bin.
  - **Behavior:** It does not depend on the threshold (absolute volume) but reacts only to the "spikes" in the sound. Use this first when you want to smooth out the texture without changing the volume.
- **Threshold (-60dB 〜 0dB) / Depth (0.0 〜 1.0):**
  - **Role:** Suppression of "unnecessary muddiness components (clusters in the band)" based on a masking model.
  - **Behavior:** Even if Threshold is set to 0.0dB, it automatically reacts if there is a "prominence that is easily audible to the ear," and Depth determines the depth of suppression. Since it does not operate during the attack phase (Onset detection), it does not crush transients.

🔪 Tonal / Transient Control

- **Tonal (-24dB 〜 +24dB):**
- Adjusts the volume of "sustained sounds" such as vocals, synth pads, and reverb tails.
- **Transient (-24dB 〜 +24dB):**
- Adjusts the volume of "momentary attack sounds" such as kick attacks and vocal consonants.

🌌 Band-specific Width Control (Stereo Expansion)

LUMINA controls the Side component using **completely different algorithms** for each frequency band.

- **Low Band:**
  - **Algorithm:** Amplitude attenuation only (pure M/S gain operation).
  - **Reason:** Manipulating the phase of the low frequency range would cause a fatal loss of kick or bass energy due to phase cancellation, so phase is absolutely protected.

- **Mid Band:**
    - **Algorithm:** M/S Matrix Control.
    - **Reason:** This is the most sensitive band where the fundamental frequencies of vocals and snares exist. Since artificial phase dispersion would make the sound pull back, only gain operations are performed to maintain clarity.
- **High Band:**
    - **Algorithm:** Schroeder All-Pass Network (series/parallel ZDF All-Pass Filter groups).
    - **Reason:** In the high frequency range, "inter-aural phase difference" is perceived as stereo spread (Air/Shimmer). When Width > 1.0, only the phase is randomly scrambled to generate an extreme stereo image that spreads beyond the speakers while maintaining full mono compatibility.

-----4. Professional Mode (Pro Mode)

Enabling Pro Mode unlocks the deep DSP parameters.

- **Oversampling (Off / 2x / 4x):**
- Increases the internal processing sample rate to eliminate aliasing noise.
- **Lookahead (0.0ms ～ 10.0ms):**
- Looks ahead for transient detection and Tame processing to prevent missing the attack.
- **Width X1 / Width X2 (20Hz ～ 20kHz):**

    - **Role:** Sets the band splitting points **completely independent** of the dynamics processing crossovers (Crossover 1/2), **purely for Width processing (spatial control)**.
    - **Use Case Example:** Allows advanced routing, such as "I want to widen the Mid band for compression to match the vocals, but I want to limit the stereo spread (Schroeder All-Pass) only to the ultra-high frequencies above 8kHz."
- **Tame Sharpness (0.1 ～ 10.0):**
- The "sharpness (Q width) of the peak" that Tame reacts to. Setting it high shaves off spikes only in an extremely narrow band (surgical), preserving the original tone.
- **Tame Speed (0.1 ～ 0.99):**
- Tame's tracking speed. Setting it fast suppresses momentary peaks, while setting it slow results in analog-like behavior.
- **Attack (0ms ～ 100ms) / Release (0ms ～ 500ms) [M/S Independent]:**
- Allows setting the Attack/Release times for the Threshold / Depth processing independently for Mid and Side.
- **HPSS Blur (0.1 ～ 2.0) / HPSS Res (3.0 ～ 65.0):**
    - **Res:** Time resolution of the median filter separating Tonal/Transient.
    - **Blur:** Blurring of the separation boundary. Lowering it results in digital glitch sounds, while raising it results in natural, smooth separation.
- **Link Amount (0.0 ～ 1.0):**
- The degree of coupling for dynamics processing between Mid and Side in M/S Mode (0.0 for dual-mono, 1.0 for full linking).

-----5. Use Cases and Recommended Workflow🛠️ Case 1: Cleaning Up Muddiness and Enhancing Clarity on the Master Bus

1. Insert LUMINA on the master track and optimize the crossovers with Auto Band.
2. Set Tame to 0.3 ～ 0.5 for all bands to smooth out subtle peaks.
3. Set the Mid band Threshold to -2.0dB and Depth to 0.4 to suppress muddiness caused by masking.
4. Turn Auto Level ON to compensate for volume changes.

🛠️ Case 2: Tight Punch-Up and Spatial Expansion on the Drum Bus

1. Set Low band Transient to +4.0dB and Tonal to -4.0dB to emphasize the kick drum's attack.
2. Open Pro Mode and set Width X2 to approximately 6000Hz.
3. Set High band Width to 1.5. Only the hi-hats and cymbals will safely spread to the left and right of the stereo field, securing space for the center vocal.

🛠️ Case 3: Creating Extreme Glitch / Smearing Textures

1. Turn Pro Mode ON and set HPSS Blur to the minimum (0.1) for a specific band.
2. Set Transient to -24dB to completely eliminate the attack, and boost Tonal by +12dB.
3. An intense smearing (reverb) texture is generated, where the attack is lost and digital artifacts are emphasized.